

# ALHASSAN JAWAD

ELECTRICAL ENGINEER • FOCUS: CYBER- & OT-SECURITY

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## ABOUT ME

A newly graduated **Electrical Engineer** (B.Sc.) with 2+ years of academic experience as a Teaching Assistant, focusing on technical evaluation and laboratory verification. During my studies, I focused on control theory and the programming of embedded systems. Currently in the final phase of getting my M.Sc. (Graduation: Q1 2027) in electrical engineering focusing on **OT-security** and **control systems**.

My background bridges electrical infrastructure with embedded systems, ranging from low-level C programming to software development. I apply **control theory**, **signal processing**, and **IoT frameworks** to engineer secure software solutions. My goal is to join a collaborative environment where I can continuously develop my skills, adapt to new challenges, and make a meaningful impact on **critical infrastructure**.

## TECHNICAL COMPETENCIES

**Programming Languages:** Python, Java, JavaScript, HTML, CSS, MATLAB, LabVIEW  
**Engineering Tools & Frameworks:** Angular, Tailwind CSS, npm, Altium Designer, PSpice, LTSpice, PostgreSQL  
**Engineering Skills:** Control Systems, Embedded Systems, System Optimization, Signal Processing, Data Analysis  
**Languages:** Swedish (Fluent), English (Fluent), Arabic (Native)

## RELEVANT EXPERIENCE

- Software Engineer (Internship) – Portabel Health

Sep 2025 – Dec 2025

Programmed and debugged system-level components for the production platform using:

  - Angular framework,
  - Tailwind CSS,
  - Node Package Manager (npm)

Executed over 130 technical tasks within an Agile development pipeline using Freedcamp, focusing on feature deployment and continuous testing across Dev and QA environments.

Managed backend data structures using PostgreSQL and translated UI/UX designs from Figma into functional code. To ensure high availability and prevent system downtime during updates, I wrote comprehensive script refactoring documentation for each new script, and created a standardized internal Git version control guideline.
- Teaching Assistant – Uppsala University

Aug 2023 – July 2025

- Instructed and evaluated 50+ students per semester on core programming & electrical concepts.
  - Was responsible for grading parts of each year’s assignments and final exam.
  - Collaborated with faculty and fellow peers to develop new course materials.
- Co-founder & Chief Technology Officer (CTO) – NordicFlip / Furrytool

March 2023 – June 2024

Startups (Closed down)

Co-founded and engineered both startups from initial conceptual design to a functional small business platform, scaling the system to support approximately 100 global buyers prior to operations closing. Managed the full-stack technical lifecycle, executing all development tasks from initial system architecture design down to code-level implementation and platform deployment.



## EDUCATION

### Master of Science (M.Sc.) in Electrical Engineering

2025 – 2027

*Uppsala University – Specialization: Control of Electrical Systems & Programming of Embedded Systems*

### Bachelor of Science (B.Sc.) in Electrical Engineering

2019 – 2022

*Department of Electrical Engineering | Uppsala University*

*Degree issued in 2026*

## SELECTED PROJECTS

### Master's Thesis: Automated Aeroponic Cultivation System

2025

*Internal Collaboration with Uppsala University*

- Programmed and deployed a rule-based MIMO (Multiple-Input Multiple-Output) control loop on an ESP32 microcontroller to measure and regulate internal temperature and humidity values.
- Integrated a real-time IoT sensor network to establish a continuous data acquisition loop, thus verifying closed-loop feedback stability and firmware execution.
- Compiled technical documentation for system initialization, debugging, and multi-variable tuning routines.

### Bachelor's Thesis: Smart Building Energy Management

2022

*External Collaboration with Studenternas IP, Uppsala*

- Wrote a custom optimization script in MATLAB to manage energy consumption and load distribution for a commercial property.
- Processed multi-variable datasets from deployed IoT sensors, isolating 71.2 kW of the facility's 185 kW total load as manageable power flexibility.
- The system verified that 38.5% of the infrastructure's peak demand could be dynamically controlled and deferred without affecting facility operations.

### Project: Miniature Elevator Prototype (Course: Project in Electronics)

2023

*Academic Project | Uppsala University*

- Helped in designing a three-level miniature elevator, handling end-to-end HIL (Hardware-In-the-Loop) integration.
- Programmed the control logic and GUI (in LabVIEW), to manage real-time level selection and state transitions while regulating motor speed profiles, to achieve a smooth acceleration while minimizing physical oscillations.

### Project: GPS Tracking Device (Course: PC-Board Construction with ECAD Tools)

2022

*Academic Project | Uppsala University*

- Designed a multi-layer GPS tracking Electronic PCB within Altium Designer.
- Integrated an ATmega328P microcontroller with a Quectel MC60 transceiver to process satellite GPS data and transmit coordinates over different networks.
- Implemented a hardware power-saving scheme by interfacing an MMA8653 accelerometer to provide system wake-up triggers based on physical motion.